INTRODUCTION

A FEW SHORT YEARS ago, artificial intelligence (AI) inhabited a small corner of the public debate. Today, following rapid advances, AI is a front-page topic at news outlets everywhere and an issue on the minds of leaders in science, business, journalism, public service, education, and politics the world over.

As we see it, both the general public and many experts in the field continue to overlook important aspects of this new age of AI. New forms of AI and human responses to them could transform nothing less than the human relationship to reality and to truth, the exploration of knowledge as well as the physical evolution of humanity, the conduct of diplomacy, and the international system. These are among the crucial issues of the coming decades, and they ought to be the guiding concerns of leaders in every arena.

The latest capabilities of AI, impressive as they are, will appear weak in hindsight as its powers increase at an accelerating rate. Powers we have not yet imagined are set to infuse our daily lives. Future systems will facilitate enormous and largely beneficial advances, improving our health while generating wealth.

But these capabilities come with technical and human risks, some of which are known and some unknown. Today's technologies already function in ways that their inventors did not predict, and this pattern is likely to continue. Each fruitful research path taken by our scientists—and there will be more than one—could yield new branches of unforeseen powers that may or may not be comprehensible or beneficial to humans.

AI seems to compress human timescales. Objects in the future are closer than they appear. As just one example, machines with the ability to define their own objectives are not far away. If we have

any hope of keeping up with the risks involved, we must respond and act within the shortest conceivable timeline. Cognizant of the stakes and the urgency of the task ahead, we note here only a few of its many facets.

As human-machine partnerships become ubiquitous, humans will have to determine these relationships' proper nature. The answers may be taken from the logic of security and efficiency, derived via the study of history, or discerned from the divine. Individuals, nations, cultures, and faiths will need to determine the limits, if any, of AI's authority over truth. They will need to decide whether to allow AI to become an intermediary between humans and reality. In this connection, they will need to choose between, on the one hand, retaining the traditional role of human enterprise (while likely ceding leadership to AI in the discovery of new knowledge) and, on the other hand, abandoning the biologically limited human mind in favor of a potentially reengineered partnership with AI on the intellectual frontier. Do we pick our objectives and harness AI to achieve them, or do we let AIs help pick some of the objectives themselves? Most urgently, humanity must give human dignity a modern and sustainable definition that can provide a philosophical orientation for our decisions in the years to come.

The advent of artificial intelligence is, in our view, a question of human survival. As we explain later in this book, AI's future faculties, operating at inhuman speeds, will render traditional regulation useless. We will need a fundamentally new form of control. For the global scientific community, the immediate task is to find technical measures for instilling intrinsic safeguards in every AI system. For their part, nations and international organizations, once they have coalesced around a consensus, must develop new political structures for monitoring, enforcement, and crisis response. That will require the resolution of not one but two "alignment problems": the technical alignment of human values and intentions with the actions of AI, and the diplomatic alignment of humans with their fellow humans.

Dr. Henry Kissinger, one of this book's coauthors, closely mentored his two collaborators on the latter problem, while they—as

previous senior leaders of Microsoft and Google—coached him on the former. Craig Mundie was Microsoft's chief technology-policy liaison to Washington and foreign governments worldwide, concurrently overseeing Microsoft Research before more recently advising the leadership of the research organization OpenAI. Eric Schmidt led Google for a decade before spending the next decade forging a network of talent and ideas at the intersection of technology, philanthropy, and security with the aim of safeguarding humanity.

The urgency of the issues that we have confronted together is such that, rather than waiting for a crisis, we deem it imperative that our society, and indeed our species, address them proactively. And, although the cause of human safety is one necessary component of a successful response to AI, it cannot answer all the questions that AI raises—for, in the age of AI, humanity will change. The question is whether and to what extent we humans will choose to continue to assert authority over how that change occurs.